

Scalable, safeguarded DOM optimization

A feasibility-first hierarchy from greedy planning to bounded QAOA proposals.

Greedy • exact recourse • exact LNS • bounded QUBO/QAOA

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Submission evidence

516

aggregate rows across 14 audited families

+1.86 pp

objective capture vs default on 100 groups

100k

generated orders: greedy + exact LNS

513 validated plans • 18 QPU jobs • 8,192 shots/job

DOM is constrained allocation

INPUT

Orders, lines, and loads

- demand, value, and service rules
- eligible DC/date alternatives
- protected cumulative ATP
- dock and optional resource limits

DECISION

Choose one outcome per order group

$$x_{gc} \in \{0,1\}$$

1 when group g selects candidate c ; 0 otherwise.

Unfilled demand is explicit and penalized.

OBJECTIVE

Maximize validated service value

fulfilled value – shipping
– thresholded unmet penalty

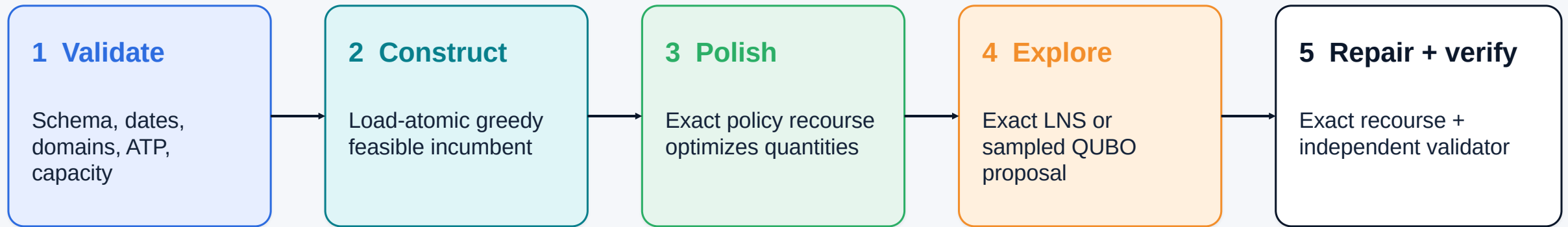
Subject to

cumulative ATP • capacity • eligibility • load cohesion • integer balance

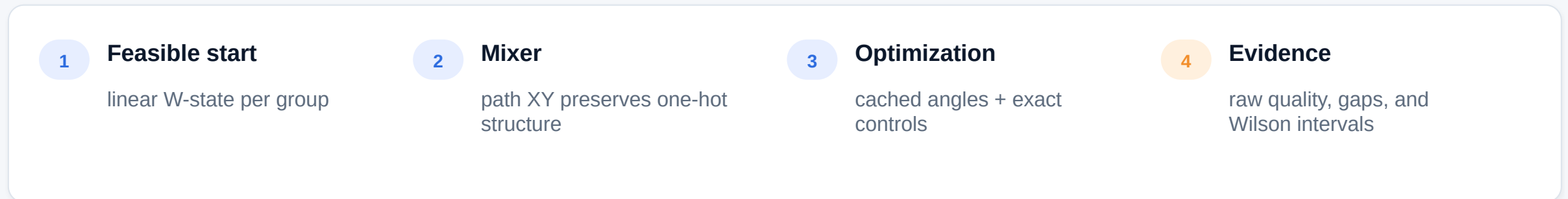
Why it is hard

Shared inventory couples groups. The solver must preserve a feasible incumbent while coordinated local search reopens only the decisions that interact.

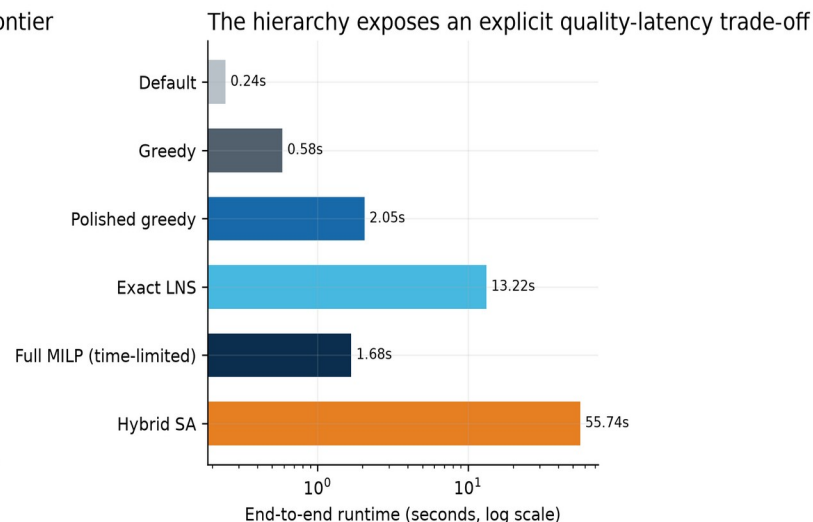
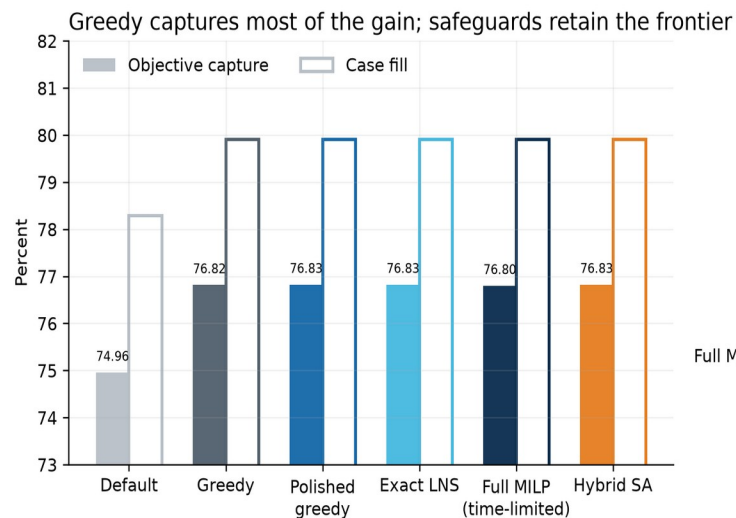
The solver hierarchy keeps feasibility deterministic



Quantum neighborhood (optional)



Greedy captures most of the gain in 0.58 s

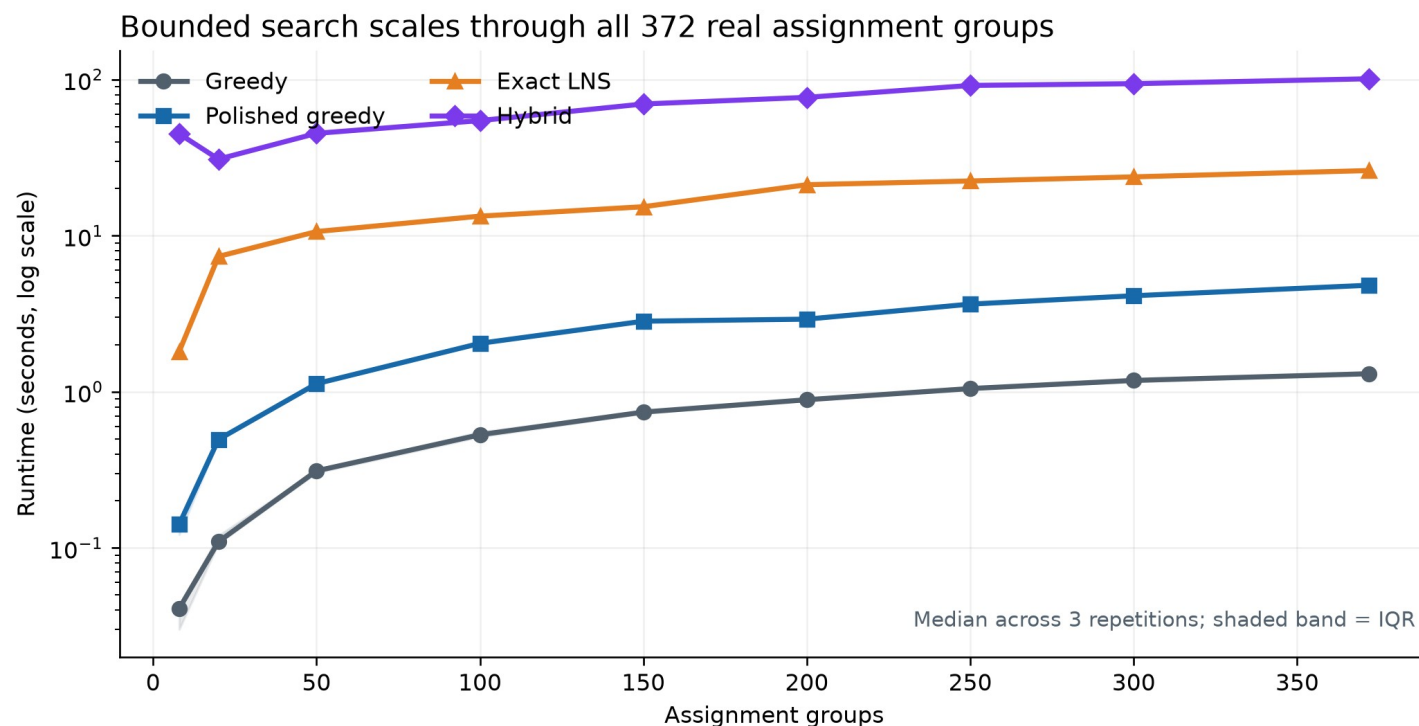
**76.83%**objective capture
after exact polish**79.91%**

case fill rate

2.05 spolished-greedy
runtime**0**validator residual
violations**+197.2 on the synthetic coordination trap**Exact LNS and QAOA proposals recover the same
coordinated gain after exact recourse.

Greedy adds +1.86 pp capture vs default; the time-limited full MILP retains a nonzero gap.

Bounded search scales through all 372 real groups

**FAST****Greedy**

1.31 s median at all 372 groups; feasible incumbent for every planning run.

QUALITY**Exact LNS**

26.14 s median and the strongest full-scope capture in the repeated study.

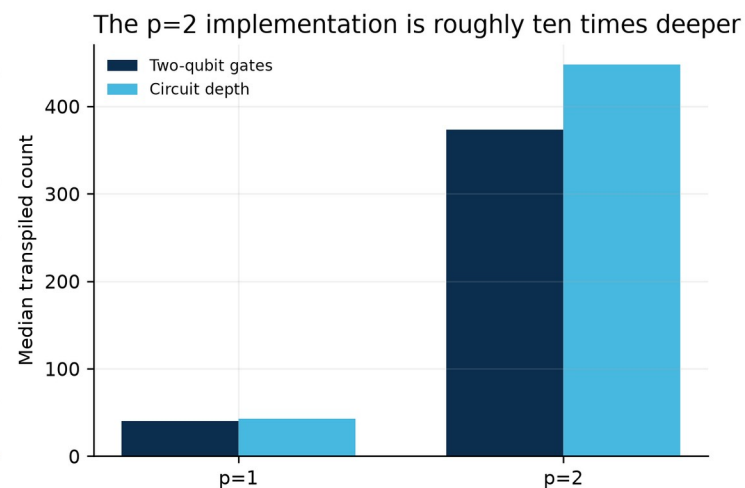
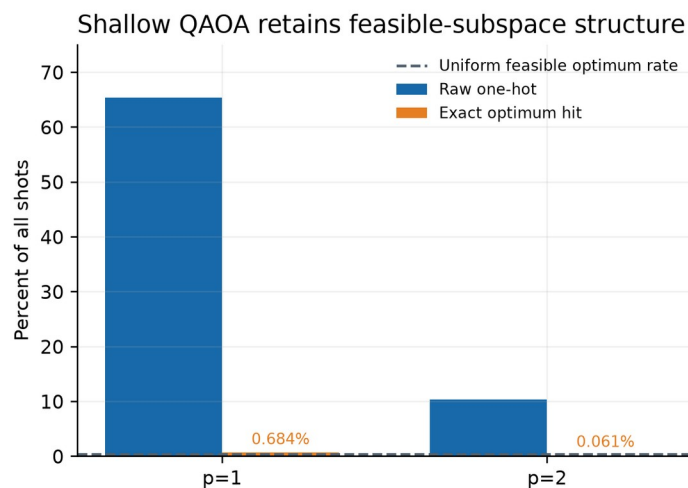
HYBRID**Fixed-width QUBO**

101.17 s median; at most 16 variables at full real scope.

Generated: greedy + exact LNS reach 100k orders; hybrid reaches 20k with a 32-variable QUBO.

Shallow QAOA better preserves feasibility

Dicke/XY-QAOA on ibm_marrakesh: 18 jobs, 8,192 shots each



p = 1

40 two-qubit gates •
depth 43

65.34%

median raw one-hot
feasibility

p = 2

373 two-qubit gates •
depth 448

10.36%

median raw one-hot
feasibility

Interpretation

p=1 exact hit: 0.684% vs 0.391% uniform control.
Exact recourse validates every final plan.

18 QPU jobs • 8,192 shots/job • runtime uses a log scale because queue outliers reach about 12,021 s.

Deploy classical now; measure QAOA

RECOMMENDATION

Production path

- FAST: release polished greedy after certification.
- QUALITY: trigger exact LNS for material conflicts.
- CERTIFY: validate every plan and record residuals.
- COPILOT: review changes, rationale, and downloads.

Improvement roadmap

NOW

Shadow-mode pilot

Confirm DCs, calendars, capacity • measure planner acceptance and realized service

NEXT

Scalable search

Adaptive LNS • rolling uncertainty • solver-engine benchmark • learned conflicts

R&D

Bounded QAOA

Shallow $p=1$ • angle transfer • sequential stopping • calibrated mitigation

FEASIBLE FIRST

SCALABLE

QAOA: BOUNDED + MEASURED

The advantage is a solver that scales now and makes every future proposal auditable.